Roll No. | | | | | | | |

EVEN END SEMESTER EXAMINATION – 202..

Name of the Program: B.Tech. Semester: VII

Name of the Course: Cryptography and Network Security Paper Code: TOE-707

Time: 3 Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks for each main question are twenty.

(iv) Each sub question carries 10 marks

Q1	(20marks)	CO1
(a)	Provide a brief explanation of the conventional encryption system, including a diagram to illustrate it.	
(b)	Differentiate cryptanalysis and cryptography. Also explain different types of cryptanalysis attacks.	
(c)	Ajay have received a secret message "MAMPRSXKMZIYT" from Ram and he knows that ram has encrypted the message using the Caeser cipher with key 4. Help Vijay to recover the original message.	
Q2	(20 marks)	CO2
(a)	(i) How many padding bits are added to a message of 140 characters if 8-bit ASCII is used for encoding and the block cipher accepts blocks of 64-bit. (ii) Why modern block ciphers are designed as combination of substitution ciphers and transposition cipher instead of transposition ciphers explain with the help of an example.	
(b)	(i) Describe the principles that govern block cipher design. Why is the use of multiple rounds essential for security. (ii) Elaborate the process of AES key Expansion in detail.	CO2
(c)	Analyze the various schemes of symmetric key distribution.	CO3
Q3	(20 marks)	CO3
(a)	Suppose that two parties A and B wish to set up a common secret key (D-H key) between themselves using the Diffie Hellman key exchange technique. They agree on 7 as the modulus and 3 as the primitive root. Party A chooses 2 and party B chooses 5 as their respective secrets. What is their D-H key.	
(b)	Evaluate the components of a public key cryptosystem. Explain how each component functions and assess their roles in securing communication. Additionally, provide a detailed description of the RSA cryptosystem.	CO4
(c)	Explain the necessity of X.509 certificate in secure communication also Describe its format.	CO4
Q4	(20 marks)	
(a)	Analyze the services offered by IPsec and evaluate its architecture in detail.	CO5

(b)	Evaluate the Kerberos authentication process, analyzing its components and how they work together to ensure secure user and service authentication in a network.	CO5
(c)	Discuss SSL session and SSL connection in detail.	CO6
Q5	(20 marks)	
(a)	Discuss the architecture of IEEE 802.11 in detail.	CO6
(b)	Analyze the methods used to achieve data protection in cloud.	
(c)	Categorize the web security attacks along with their countermeasures.	